1 Pairwise Correlation with Missing Data

a. **Calculate the pairwise correlation matrix.**

See Table 1.

Table 1: Pairwise correlation matrix

	x_1	x_2	x_3	x_4	x_5
x_1	1.0000	0.3328	0.4185	0.2553	0.7323
x_2	0.3328	1.0000	0.2510	0.2877	0.7875
x_3	0.4185	0.2510	1.0000	0.4836	0.6843
x_4	0.2553	0.2877	0.4836	1.0000	0.7826
x_5	0.7323	0.7875	0.6843	0.7826	1.0000

b. **Calculate the variance of each series, using every observation that series has.**

See Table 2.

Table 2: Variance of each series (all observed values)

Series	n_{obs}	Variance
x_1	41	4.060739×10^{-4}
x_2	41	4.800663×10^{-4}
x_3	36	1.511102×10^{-4}
x_4	41	8.417388×10^{-4}
x_5	42	2.731496×10^{-4}

c. **Is the correlation matrix positive semi-definite? Show how you know.**

It is not PSD. There correlation matrix has a negative eigenvalue, so the matrix cannot be PSD.

Eigenvalues of the correlation matrix: [-0.089279, 0.461340, 0.719085, 0.823437, 3.085416]

d. **If it is not, use Higham's method to find the nearest positive semi-definite correlation matrix. Report the matrix and its eigenvalues. Which entries changed the most?**

Since it is not PSD, we use Higham's method to find the nearest PSD matrix. See Table 3 for the matrix and Table 4 for the eigenvalues. We can see that it changes the smallest value to approximately 0 and altered the other eigenvalues slightly as a tradeoff.

The entries involving x_5 changed the most, substantially more than the other series. This was manually calculated using an LLM given the results of my output in problem1.py.

Table 3: Higham nearest PSD correlation matrix

	x_1	x_2	x_3	x_4	x_5
x_1	1.0000	0.3495	0.4259	0.2708	0.6946
x_2	0.3495	1.0000	0.2606	0.3075	0.7393
x_3	0.4259	0.2606	1.0000	0.4925	0.6626
x_4	0.2708	0.3075	0.4925	1.0000	0.7377
x_5	0.6946	0.7393	0.6626	0.7377	1.0000

Iterations = 36, converged = True, $\|C_{\text{psd}} - C\|_F = 0.121427$

Table 4: Eigenvalues of the Higham PSD correlation matrix (ascending)

	Eigenvalue
λ_1	−0.000000
λ_2	0.450082
λ_3	0.703476
λ_4	0.806708
λ_5	3.039734

Table 5: Change in correlations, $\Delta = C_{\text{Higham}} - C_{\text{pairwise}}$

	x_1	x_2	x_3	x_4	x_5	$\sum_j \lvert\Delta_{ij}\rvert$	$\sqrt{\sum_j \Delta_{ij}^2}$
x_1	0	0.0167	0.0074	0.0155	−0.0377	0.0773	0.0447
x_2	0.0167	0	0.0096	0.0198	−0.0482	0.0943	0.0556
x_3	0.0074	0.0096	0	0.0089	−0.0217	0.0476	0.0264
x_4	0.0155	0.0198	0.0089	0	−0.0449	0.0891	0.0522
x_5	−0.0377	−0.0482	−0.0217	−0.0449	0	**0.1525**	**0.0789**

$\|\Delta\|_F = 0.1214$

e. **Combine your results into a covariance matrix. Report it.**

See Table 5.

Table 6: Covariance matrix (Higham correlation, full-sample variances)

	x_1	x_2	x_3	x_4	x_5
x_1	4.060739e-04	1.542911e-04	1.055115e-04	1.583111e-04	2.313238e-04
x_2	1.542911e-04	4.800663e-04	7.018913e-05	1.954867e-04	2.677277e-04
x_3	1.055115e-04	7.018913e-05	1.511102e-04	1.756475e-04	1.346081e-04
x_4	1.583111e-04	1.954867e-04	1.756475e-04	8.417388e-04	3.537164e-04
x_5	2.313238e-04	2.677277e-04	1.346081e-04	3.537164e-04	2.731496e-04

2 VaR and ES for a Single Stock

a. **What is your VaR at the 95% and 99% levels, in percent? Explain your method.**

The VaR is 3.2245% at 95% is and 4.4605% at 99%. Given that the arithmetic returns of SPY and ϵ are normally distributed, I could use delta normal and historical to calculate VaR. However, I went with delta normal because there is limited history. With 501 observations, 1% tail estimates rely on just 5 observations.

Mimicking a similar approach to Assignment 2, Question 5, my method is the following:
Calculate returns and run the regression.

Table 7: OLS regression of r_{XYZ} on r_{SPY} (arithmetic returns, $n = 500$)

Parameter	Estimate
α	−0.000304
β	1.193311
Residual sd $(n-1)$	0.014590
Residual se $(n-2)$	0.014605
R^2	0.446083

Calculate the sample covariance matrix.

Table 8: Sample covariance matrix of arithmetic returns

	XYZ	SPY
XYZ	3.842986×10^{-4}	1.436584×10^{-4}
SPY	1.436584×10^{-4}	1.203864×10^{-4}

Table 9: Delta normal VaR (100 shares XYZ, value $7,451.83)

Level	VaR (%)
95%	3.2245
99%	4.5605

We can also look at Historical VaR.

Table 10: 1-day VaR of XYZ, arithmetic returns (percent)

Level	Historical VaR (%)	Normal VaR (%)
95%	3.1881	3.2245
99%	4.0760	4.5605

Table 11: First four moments of arithmetic returns ($n = 500$)

	XYZ	SPY
Mean	3.108624×10^{-18}	2.547717×10^{-4}
Variance	3.842986×10^{-4}	1.203864×10^{-4}
Skewness	0.208302	0.014383
Excess kurtosis	−0.135018	0.077080

b. **What is your ES at the 95% and 99% levels, in percent? Explain your method.**

ES is 4.0436% at the 95% level and 5.2248% at the 99% level. For similar reasons as VaR, delta normal values were used to determine ES. There is quite a large difference between delta normal and historical ES, suggesting that the distribution of the data has thinner/more skewed tails than in normal distribution.

Table 12: Delta normal ES (percent)

Level	ES (%)
95%	4.0436
99%	5.2248

Table 13: Historical ES (percent)

Level	ES (%)
95%	3.7986
99%	4.5158

3 Copula VaR and ES for a Two Book Portfolio

a. **For each stock, fit a normal, a Student's t, and a normal inverse Gaussian distribution to its returns. Report the fitted parameters. Which distribution fits each stock best? Justify your choice.**

Best fit for each stock, determined by lowest AICc (See Tables 13-15):

- S1: Normal
- S2: Student's t
- S3: NIG
- S4: Normal
- S5: Student's t
- S6: NIG
- S7: Normal
- S8: Student's t
- S9: NIG
- S10: Student's t

Table 14: Normal fits (arithmetic returns)

Stock	μ	σ	Log-lik	AICc
S1	0.000904	0.011826	3019.0026	−6033.9932
S2	0.000800	0.015348	2758.3587	−5512.7054
S3	0.000424	0.018898	2550.2622	−5096.5124
S4	0.000780	0.009933	3193.4927	−6382.9735
S5	0.000709	0.018951	2547.4534	−5090.8947
S6	0.000805	0.015085	2775.6075	−5547.2030
S7	0.000693	0.013674	2873.8529	−5743.6938
S8	0.000374	0.017112	2649.5592	−5295.1064
S9	0.001746	0.020647	2461.7588	−4919.5057
S10	0.000165	0.013061	2919.6681	−5835.3242

Table 15: Student's t fits (arithmetic returns)

Stock	μ	σ	ν	Log-lik	AICc
S1	0.000906	0.011767	200.0000	3018.9470	−6031.8700
S2	0.000581	0.011672	4.8577	2816.6919	−5627.3597
S3	0.002291	0.013657	4.1605	2622.2436	−5238.4631
S4	0.000782	0.009883	200.0000	3193.3140	−6380.6038
S5	0.000833	0.015577	6.2893	2578.4468	−5150.8695
S6	0.002445	0.010655	3.8237	2847.3821	−5688.7400
S7	0.000696	0.013605	200.0000	2873.6837	−5741.3434
S8	0.000654	0.011424	3.4833	2749.6558	−5493.2876
S9	0.003383	0.015310	4.3852	2521.3933	−5036.7625
S10	0.000405	0.009427	4.1213	2990.4406	−5974.8570

Table 16: Normal inverse Gaussian fits (arithmetic returns)

Stock	μ	α	β	δ	Log-lik	AICc
S1	0.617908	16686.6421	−13747.7589	0.424453	3019.1365	−6030.2328
S2	−0.000305	73.2063	4.8635	0.016593	2815.0516	−5622.0630
S3	0.010474	79.8374	−36.3448	0.019657	2644.8406	−5281.6410
S4	0.348663	9722.8424	−6673.6027	0.368590	3193.5989	−6379.1576
S5	0.001497	73.5767	−2.2321	0.025959	2577.1249	−5146.2095
S6	0.008272	91.6816	−41.6848	0.014628	2872.7178	−5737.3955
S7	0.240255	3374.0568	−1767.6007	0.389511	2873.9922	−5739.9442
S8	0.001183	50.6226	−2.8875	0.014162	2747.5672	−5487.0941
S9	0.010128	63.1595	−22.9266	0.021515	2536.7749	−5065.5096
S10	0.001048	76.6811	−5.3661	0.012595	2989.3710	−5970.7018

b. **Using the distributions you chose in part (a), fit a Gaussian copula and a Student's t copula. Which one fits better? Justify your choice.**

The Gaussian has a lower AICc and BIC than the t copula. It also has a higher log-likelihood. By all criterion, the Gaussian fits better.

Table 17: Copula fits on $U_i = F_i(r_i)$, margins from part (a)

Copula	ν	Log-lik	k	AICc	BIC
Gaussian	–	2110.0225	45	−4125.7053	−3909.1959
Student's t	200.0000	2109.5824	46	−4122.6276	−3901.4081

c. **Using the copula you chose in part (b), simulate the portfolio with at least 5,000 draws. Calculate the absolute VaR and ES at the 95% level, in dollars, for each stock, each book, and the total portfolio. Present the results in a table. Discuss the results.**

As seen in the problem3_holdings.csv, the first five stocks are in Book A and the last five are in Book B. Consequently, since Book B had stocks with relatively higher VaR and ES, Book B overall also had higher values than Book A. In Book A there are a total of 1900 shares whereas Book B only has 1375.

Looking at VaR, Book A at minimum could lose on a normal 5% bad day could lose \$3482.08 whereas for Book B, it could lose \$5336.31. Looking at ES, the expected loss given a loss beyond 5% VaR is \$4663.61 for Book A and \$7565.29 for Book B. Based on these risk metrics, Book A seems less risky than Book B.

Combining the two books does seem to lower the risk, suggesting subadditivity holds here for both VaR and ES. $3482.08 + 5336.31 \geq 7189.31$ and $4663.61 + 7565.20 \geq 10171.80$, suggesting that diversifying/combining the books does diminish the total portfolio risk.

Table 18: Gaussian copula simulation (5,000 draws): 95% absolute VaR and ES ($)

	VaR ($)	ES ($)
S1	878.42	1,122.05
S2	1,044.28	1,488.03
S3	789.06	1,173.68
S4	647.54	828.02
S5	1,053.02	1,460.39
S6	1,182.78	1,803.12
S7	865.93	1,080.85
S8	635.62	999.15
S9	3,340.31	4,952.18
S10	482.76	689.04
Book A	3,482.08	4,663.61
Book B	5,336.31	7,565.29
Total	7,189.31	10,171.80